Meat also includes the glands and organs of animals.

4.1.2 Structure of meat

Animal flesh consists of muscle tissue or fibres, connective tissue and fatty (adipose) tissue. Lean meat is the muscle tissue of animals.

Meat muscle is made up of bundles of muscle fibres held together by creamy white connective tissues. Tendons join muscles (made up of bundles of muscle fibres, surrounded by connective tissue) to the bones of animals.

Muscle cells comprises of

1. Water
2. Minerals
3. Fat
4. Proteins
5. Vitamins
6. The red pigment called myoglobin

Fat: Fat is distributed throughout meat in small particles or in large masses. The pattern formed by the uniform distribution of fat in small "lakes" throughout the muscle or lean flesh is called **"marbling"** and is considered an important factor in contributing tenderness and flavour to muscle tissue

The colour of meat

The colour of meat is due to the red pigment called **'myoglobin'**.



Fig 4.2: Marbling in meat

A large amount of the iron found in meat is in the liver, an organ constituting only a small portion of the carcass.

Offals (Organ meats)

Offal also called as organ meats, refers to the internal organs and entrails of a butchered animal. The name offal means "off fall", in other words, the bits which fall from an animal when it is butchered. The term offals generally covers organs such as the heart, liver and lungs (collectively known as the pluck) and other organs like the kidneys, brains, head, feet, tongue, intestines and tails. Offal from birds is generally referred to as giblets.

Offals are a highly nutritious food and an excellent source of protein. The liver, kidney and heart are a good source of iron and vitamins A, and D.



Fig 4.3: Organ meats- Liver, Brain, Heart and Kidney

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